

FlowPilot manuscript: paragraph-by-paragraph storyline

One bullet per scientific paragraph, in manuscript order. P01-P56 exclude captions, author information and references.

Abstract

- **P01** Overview: research assistant, architecture, benchmark and experimental evidence.

Introduction

- **P02** Motivation: flow opportunities require laboratory-specific process design.
- **P03** Problem: chemistry, stage conditions and equipment decisions are coupled.
- **P04** Context: planning, simulation and optimization support different development tasks.
- **P05** Context: scientific agents connect researcher intent to tools and operations.
- **P06** Prior work: existing flow and process agents define the comparison.
- **P07** Design requirement: unite chemistry, quantitative conditions and available hardware.
- **P08** Contribution: FlowPilot supports the researcher through end-to-end process design.
- **P09** Mechanism: retrieval, council and calculations share one inspectable design record.
- **P10** Evidence roadmap: retrieval, architecture comparison and two connected reactions.

Results: Architecture and Knowledge (Figures 1-3)

- **P11** Define the problem: intake connects chemistry, objectives and laboratory constraints.
- **P12** Coordinate decisions: specialists review alternatives and recalculate revised conditions.
- **P13** Realize the proposal: compatible equipment, stage data and topology agree.
- **P14** Ground chemical choices: corpus coverage and annotated process precedents.
- **P15** Ground engineering choices: rule coverage, applicability and calculation boundaries.
- **P16** Retrieve evidence: chemistry-aware queries and progressively relaxed search filters.
- **P17** Measure search changes: enriched queries and observed reranking activity.
- **P18** Interpret retrieval: better metadata alignment is not transferable kinetic evidence.

Results: Architecture Comparison and Experimental Transition (Figure 4)

- **P19** Test architecture: matched one-shot comparisons across three complementary chemistries.
- **P20** Define assessment: model diversity, repeats, fixed criteria and three judges.
- **P21** Explain fairness: matched information, unequal computation and repeat-level statistics.
- **P22** Interpret gains: model-dependent improvement and competitive local-model designs.
- **P23** Assess defects: fewer judge flags, distinguished from confirmed physical failures.
- **P24** Locate benefits: process integration, chemistry-specific effects and generation costs.
- **P25** Interpret ablations: council matters; additional refinement is not uniformly beneficial.
- **P26** Assess exploration: candidate budgets, engineering profiles and selection variability.
- **P27** Connect to practice: GUI provenance, inspectable disagreements and remaining defects.
- **P28** Introduce experiments: three priority sets, two realized configurations per chemistry.

Results: Connected-Process Experiments (Figures 5-6)

- **P29** Giese purpose: separate radical addition from oxygen-dependent sulfoxide formation.
- **P30** Giese realization: compatible irradiation, feeds and explicitly defined time indices.
- **P31** Giese learning: backflow feedback, revised oxygen delivery and measured yields.
- **P32** Amidation purpose: couple thioester formation, amine addition and downstream conversion.
- **P33** Amidation realization: preserve stoichiometry and account for cumulative downstream flow.
- **P34** Amidation learning: relate yield differences to stage allocation and flow.

Discussion

- **P35** Synthesize novelty: the connected, laboratory-specific process is the design target.
- **P36** Interpret architecture: coordinated review matters more than agent count alone.
- **P37** Interpret experiments: apparatus behavior and complete-sequence performance require measurement.
- **P38** Define progression: traceable decisions and measured feedback support prospective refinement.

Conclusion

- **P39** Restate contribution: one researcher-facing environment links knowledge, design and hardware.
- **P40** Consolidate evidence: benchmark gains and experimentally implemented connected reactions.
- **P41** Take-home message: support experimental development rather than replace experimentation.

Methods

- **P42** Inputs: fixed questions, frozen packages, evidence authority and campaign separation.
- **P43** Knowledge: corpus annotation, missing metadata and handbook-derived rules.
- **P44** Live retrieval: filters, weighting, relaxation and source exclusion.
- **P45** Offline analysis: embedding comparisons, query sampling and alignment definitions.
- **P46** Liquid calculations: molar feeds, stage balances and residence-time definitions.
- **P47** Gas/thermal calculations: reference states, time bases and screening assumptions.
- **P48** Council procedure: candidate generation, specialist review, revisions and selection records.
- **P49** Inventory procedure: form/document input, validation and equipment-capability checks.
- **P50** Output procedure: synchronized rendering, autosave, feedback and reconstruction requirements.
- **P51** Benchmark generation: matched inputs, models, repeats and computation differences.
- **P52** Judging: masked packets, fixed ratings, applicability and equal-weight aggregation.
- **P53** Statistics: repeat-level means/SD, flag aggregation and cost definitions.
- **P54** Ablation procedure: module conditions, case variation and separate historical study.
- **P55** Experimental procedure: sample loops, staged feeds, equipment and operating conditions.
- **P56** Analysis: NMR basis, characterization, grouped results and outstanding metadata.